



Clinical Dosing Guide

GutGuard SynBIOTIC+ · Three-Tier Protocol · LCA Reference

For LCA use only. This guide supports your clinical judgment when recommending GutGuard SynBIOTIC+ within FDA CPR No. FR-40000015571456 claims. Not a prescription; not a substitute for individualized clinical assessment.

Three dosing tiers, matched to patient state: Maintenance (1-1), Active management (2-2), Peak protocol (2-2-2).

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1. The Three-Tier Dosing Protocol

All capsules taken with water on an empty stomach (or at least 30 minutes before food).

Capsules are gastro-resistant; opening them is not recommended.

Tier	Daily Dose	Indication	Recommended Package
Maintenance 1–1	1 capsule AM (on waking, empty stomach) 1 capsule PM (30 min before bedtime)	Asymptomatic patient. No diagnosed conditions. Wellness / preventive use.	Start (40 caps = 20 days) or Grow (120 caps = 60 days)
Active management 2–2	2 capsules AM (on waking, empty stomach) 2 capsules PM (30 min before bedtime)	Patient experiencing or reporting symptoms: bloating, irregular bowel patterns, fatigue, mild inflammatory markers.	Grow (120 caps = 30 days) or Peak (400 caps = 100 days)
Peak protocol 2–2–2	2 capsules AM (on waking, empty stomach) 2 capsules midday (before lunch) 2 capsules PM (30 min before bedtime)	Active diagnosed conditions. Patient committed to full 90-day Anti-Aging Therapy Program (ATP).	Peak (400 caps ≈ 67 days at 6/day, supplement to complete 90-day course)

Timing rationale: Empty-stomach AM dosing maximizes probiotic survival to the colon (less competition with food-induced gastric acid). PM dosing aligns with the circadian peak of intestinal cell turnover (the gut's repair window). The midday dose in Peak provides continuous substrate availability for sustained MBS activation.

2. Common Patient Scenarios

These archetypes are starting points. Always individualize based on your clinical assessment.

Patient Presentation	Suggested Tier	Duration & Counseling Notes
Wellness-oriented adult, asymptomatic, preventive use	1–1	Minimum 60 days to assess subjective response. Set expectations: maintenance use shows gradual benefit; no acute changes are typical.
Mild functional GI complaints: bloating, irregular bowel patterns, post-meal discomfort	2–2	Start with 30-day trial. Many patients report improvement in stool consistency and bloating within 2–3 weeks. If no response by day 30, consider Peak tier or refer for further workup.
Post-antibiotic dysbiosis (recent broad-spectrum course)	2–2	Begin 2–3 days after antibiotic course completion. Minimum 30 days; ideally 60 days to restore microbial diversity. Space doses at least 2 hours from any residual antibiotic.
Metabolic syndrome features: dyslipidemia, insulin resistance, central adiposity	2–2 → 2–2–2	Start at 2–2 for first 30 days; escalate to Peak (2–2–2) if patient committed and tolerating. Pair with standard metabolic management. Reassess inflammatory markers at 60 and 90 days.
Chronic fatigue with suspected gut involvement	2–2–2	Full 90-day Peak protocol. Many patients report energy improvement after the 30–45 day mark as mitochondrial pathway activation establishes. Track subjective energy scores weekly.
Inflammatory conditions (under specialist co-management)	2–2–2	Adjunct only — not a replacement for primary therapy. Coordinate with treating specialist. Full 90-day Peak protocol. Document baseline and follow-up inflammatory markers.
Peri- and post-menopausal women	2–2	Estrobolome support pathway. 60–90 day course. Often paired with patient-driven lifestyle changes. Reassess at 60 days.
Aesthetic / anti-aging focus, otherwise healthy	2–2–2	Full 90-day Peak ATP. The Urolithin-A + L-Tryptophan combination targets mitochondrial renewal pathways most active under sustained dosing. Skin and energy outcomes typically reported 45–60 days in.

3. Contraindications and Precautions

Do not recommend without specialist consultation

- Severely immunocompromised patients (active chemotherapy, post-transplant on immunosuppression, advanced HIV with low CD4).

- Central venous catheter in place, or recent GI surgery with anastomosis healing within 4 weeks.
- Acute pancreatitis, active GI bleeding, or short bowel syndrome.
- Pregnancy and lactation — GutGuard SynBIOTIC+ is not contraindicated in pregnancy, but the LCA program does not currently recommend this population pending dedicated review.

Use with caution / consider dose adjustment

- Patients on warfarin or other vitamin K-sensitive anticoagulation — some probiotic strains influence vitamin K production. Monitor INR more closely during first 30 days.
- Severe renal impairment — review the L-Tryptophan content with the patient's nephrologist.
- Active SIBO (small intestinal bacterial overgrowth) — may transiently worsen symptoms in the first 1–2 weeks. Start at 1–1 and titrate up only if tolerated.

Drug-supplement timing

- Levothyroxine: take SynBIOTIC+ at least 4 hours away.
- Bisphosphonates: take at least 2 hours away.
- Antibiotics (any class): take SynBIOTIC+ at least 2 hours away. Probiotic doses too close to antibiotics reduce viable strain delivery.
- Iron supplements: take at least 2 hours away to avoid tannin/mineral binding.

4. What to Tell Patients About the Timeline

Setting accurate expectations is the single most important thing you can do for adherence. The formulation works through three sequential mechanisms; each has its own response window.

Phase	Mechanism Active	What the Patient May Notice
Days 1–14 Repair	Gut environment restoration: pre + probiotics colonize and rebalance the intestinal flora.	Possible mild transient changes in stool pattern, occasional bloating as flora rebalances. Usually resolves within 7–10 days.
Days 14–45 Manage	Systemic inflammation modulation: postbiotic metabolites (SCFAs) signal the gut-immune axis.	Improved digestive comfort, more regular bowel patterns, sometimes early energy improvement. Skin clarity changes occasionally noted.
Days 45–90 Regenerate	MBS activation: Urolithin-A + L-Tryptophan drive mitophagy and	Sustained energy, improved sleep architecture, mood regulation, skin texture changes. The Peak protocol's primary value emerges in this window.

	serotonin-pathway support.	
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Counseling tip: Patients who expect overnight results stop early. Tell them the first 2 weeks are recalibration, weeks 2–6 are when GI improvements typically show, and weeks 6–12 are when the energy and regeneration benefits emerge.

5. Adverse Event Management

Most reactions are transient and resolve with dose adjustment. Always document and report serious events through the Doctor Portal.

Reaction	First-line Response	Escalation
Mild bloating or gas in first 7–10 days	Continue current dose; expect resolution by day 14. Increase water intake.	Persists past day 21: reduce to 1–1 for 7 days, then re-escalate.
Loose stools	Reduce to 1–1 for 7 days. Usually resolves.	If associated with fever, blood, or persists past 7 days at reduced dose: discontinue and assess for infectious cause.
Constipation	Increase water intake, add 1–2 g fiber daily. Maintain dose.	Persists past 14 days: temporary dose reduction to 1–1 and reassess.
Headache or fatigue in first 2 weeks	Usually a transient die-off (Herxheimer-like) response. Maintain dose, increase hydration.	Severe or persists past day 14: reduce to 1–1. Discontinue if not resolving.
Rash, urticaria, or any allergic-type reaction	Discontinue immediately. Standard allergy management.	Report through Doctor Portal as a serious adverse event. Do not resume.
Any systemic infection signs (fever + new GI symptoms)	Discontinue. Full clinical workup.	Immediate escalation to Medical Director (Dr. Shane Animas). Portal report within 24h.

6. Mechanism of Action and Supporting Literature

This section provides the scientific basis for the GutGuard SynBIOTIC+ formulation. Citations are to peer-reviewed literature on the mechanisms and active components. Clinical evidence specific to the GutGuard formulation is being generated through the 90-day LCA pilot — cite these foundational works when discussing the product with patients or colleagues.

6.1 The Gut-Mitochondrial Axis

The formulation is built on the emerging understanding that the gut microbiome and host mitochondrial function are bidirectionally regulated. Short-chain fatty acids (SCFAs) produced by gut bacteria — particularly butyrate — directly influence mitochondrial biogenesis through PGC-1 α signaling. Conversely, mitochondrial dysfunction impairs intestinal barrier integrity and shifts microbial composition.

- **Clark A, Mach N.** The gut-mitochondria axis: a new perspective for inflammation prevention. *Frontiers in Physiology* 2017;8:319.
- **Saint-Georges-Chaumet Y, Edeas M.** Microbiota-mitochondria inter-talk: consequence for microbiota-host interaction. *Pathogens and Disease* 2016;74(1):ftv096.
- **Bajpai P, Darra A, Agrawal A.** Microbe-mitochondrion crosstalk and health. *Mitochondrion* 2018;39:20–25.

6.2 Pre + Pro + Postbiotic Synergy

The 80 billion CFU / 17-strain probiotic load is designed for breadth of colonization rather than concentration of any single strain. Prebiotic fiber substrates support engraftment; postbiotic metabolites are pre-formed signaling molecules that act independent of live colonization — important for immunocompromised or partially compromised patients.

- **Salminen S, Collado MC, Endo A, et al.** The International Scientific Association of Probiotics and Prebiotics consensus statement on the definition and scope of postbiotics. *Nature Reviews Gastroenterology & Hepatology* 2021;18(9):649–667.
- **Hill C, Guarner F, Reid G, et al.** Expert consensus document: The International Scientific Association for Probiotics and Prebiotics consensus statement on the scope and appropriate use of the term probiotic. *Nature Reviews Gastroenterology & Hepatology* 2014;11(8):506–514.
- **Gibson GR, Hutkins R, Sanders ME, et al.** Expert consensus document: The International Scientific Association for Probiotics and Prebiotics consensus statement on the definition and scope of prebiotics. *Nature Reviews Gastroenterology & Hepatology* 2017;14(8):491–502.

6.3 Urolithin-A and Mitophagy

Urolithin-A is a gut-microbiota-derived metabolite of ellagitannins. It is the most-studied small-molecule inducer of mitophagy — the selective autophagy of damaged mitochondria — in humans. The Andreux 2019 study established human safety and biomarker effects; the Liu 2022 study demonstrated functional benefit on muscle endurance in middle-aged adults.

- **Andreux PA, Blanco-Bose W, Ryu D, et al.** The mitophagy activator urolithin A is safe and induces a molecular signature of improved mitochondrial and cellular health in humans. *Nature Metabolism* 2019;1(6):595–603.
- **Liu S, D'Amico D, Shankland E, et al.** Effect of urolithin A supplementation on muscle endurance and mitochondrial health in older adults: a randomized clinical trial. *JAMA Network Open* 2022;5(1):e2144279.

- **Ryu D, Mouchiroud L, Andreux PA, et al.** Urolithin A induces mitophagy and prolongs lifespan in *C. elegans* and increases muscle function in rodents. *Nature Medicine* 2016;22(8):879–888.
- **Singh A, D'Amico D, Andreux PA, et al.** Urolithin A improves muscle strength, exercise performance, and biomarkers of mitochondrial health in a randomized trial in middle-aged adults. *Cell Reports Medicine* 2022;3(5):100633.

6.4 L-Tryptophan and the Gut-Brain Axis

L-Tryptophan is the precursor for both serotonin (95% of which is synthesized in the gut) and melatonin. Gut microbiota directly modulate tryptophan metabolism through the kynurenine and indole pathways, affecting mood, sleep, and intestinal barrier function. The L-Tryptophan inclusion in the formulation supplies substrate while the probiotic component supports the conversion machinery.

- **Roager HM, Licht TR.** Microbial tryptophan catabolites in health and disease. *Nature Communications* 2018;9(1):3294.
- **Agus A, Planchais J, Sokol H.** Gut microbiota regulation of tryptophan metabolism in health and disease. *Cell Host & Microbe* 2018;23(6):716–724.
- **Kennedy PJ, Cryan JF, Dinan TG, Clarke G.** Kynurenine pathway metabolism and the microbiota-gut-brain axis. *Neuropharmacology* 2017;112(Pt B):399–412.

6.5 Probiotic Effects on Inflammatory Markers

Multi-strain probiotic formulations have reproducible effects on systemic inflammatory markers, particularly CRP and TNF- α , in patients with baseline elevation. The effect size in healthy controls is small; the effect in metabolic syndrome, IBS, and other inflammation-driven conditions is consistently larger.

- **Milajerdi A, Mousavi SM, Sadeghi A, et al.** The effect of probiotics on inflammatory biomarkers: a meta-analysis of randomized clinical trials. *European Journal of Nutrition* 2020;59(2):633–649.
- **Mazidi M, Rezaie P, Ferns GA, Vatanparast H.** Impact of probiotic administration on serum C-reactive protein concentrations: systematic review and meta-analysis of randomized control trials. *Nutrients* 2017;9(1):20.

6.6 The Gut-Mitochondria-Aging Connection (ATP rationale)

Mitochondrial dysfunction is a recognized hallmark of aging (López-Otín et al. 2013, updated 2023). Restoring mitochondrial quality control — through enhanced mitophagy — is one of the few interventions with cross-species evidence of healthspan extension. This is the rationale for the 90-day Peak protocol's framing as an Anti-Aging Therapy Program (ATP).

- **López-Otín C, Blasco MA, Partridge L, et al.** Hallmarks of aging: An expanding universe. *Cell* 2023;186(2):243–278.
- **Sun N, Youle RJ, Finkel T.** The mitochondrial basis of aging. *Molecular Cell* 2016;61(5):654–666.

7. GutGuard-Specific Evidence Status

Be honest with patients and colleagues about the evidence base. The literature above supports the **components and mechanisms** of GutGuard SynBIOTIC+. Clinical evidence specific to the GutGuard formulation itself is being generated through this program.

- **Formulation development:** MERAV (ISO-certified bio-molecular manufacturing) in collaboration with MSU-IIT and three other academic institutions. CFU stability, strain identification, and manufacturing controls documented.
- **Regulatory status:** FDA-registered as a food supplement under CPR No. FR-40000015571456. Claims must remain within registered scope — supports gut health, supports the gut-mitochondrial axis. No disease-treatment claims.
- **Clinical evidence (in progress):** The 90-day LCA pilot is the structured clinical data collection phase. Documented outcomes will be presented at PCLM and PNA, and where appropriate, submitted to peer-reviewed venues. LCAs are co-authorship eligible.
- **Adverse event tracking:** Real-world adverse event data is collected through the Doctor Portal. Aggregate reports available to LCAs upon request from the Medical Director.

8. Talking to Patients: Within-Claims Language

GutGuard SynBIOTIC+ is a registered food supplement, not a drug. Stay within mechanism-based language. The patient hears “supports” — not “treats,” “cures,” or “prevents.”

Safe Language	Avoid
“Supports gut health and the gut-mitochondrial axis.”	“Treats your IBS / leaky gut / inflammation.”
“Supports cellular renewal and energy production.”	“Will cure your fatigue / make you younger.”
“May support healthy bowel function during recovery.”	“Will fix your gut problem.”
“Research on the active components is promising; we are gathering our own data through this program.”	“Clinically proven” or “FDA-approved to treat” (the FDA registration is for the supplement, not for treatment claims).

Reminder: Disclose your LCA affiliate status to patients. “I’m a GutGuard Lead Clinical Adopter” is the simplest, cleanest disclosure and meets PMA ethics expectations.